

1. Starting from 0, the cheapest path possible connecting 0 and a new vertex is $(0, 7)$ (weight 1)

→ Add $(0, 7)$, vertex 7 and 1 weight

Similarly, start the cheapest path possible connecting 0 or 7 and a new vertex is $(0, 4)$ (weight 5)

→ Add $(0, 4)$, vertex 4 and 5 weight

Repeat the process step by step, we add:

$(4, 1)$, 1 and 5 weight

$(1, 2)$, 2 and 5 weight

$(2, 5)$, 5 and 3 weight

$(2, 6)$, 6 and 6 weight

$(6, 3)$, 3 and 2 weight

$(3, 10)$, 10 and 4 weight

$(10, 13)$, 13 and 4 weight

$(13, 12)$, 12 and 1 weight

$(12, 16)$, 16 and 1 weight

$(16, 9)$, 9 and 5 weight

~~$(13, 17)$, 17 and 7 weight~~

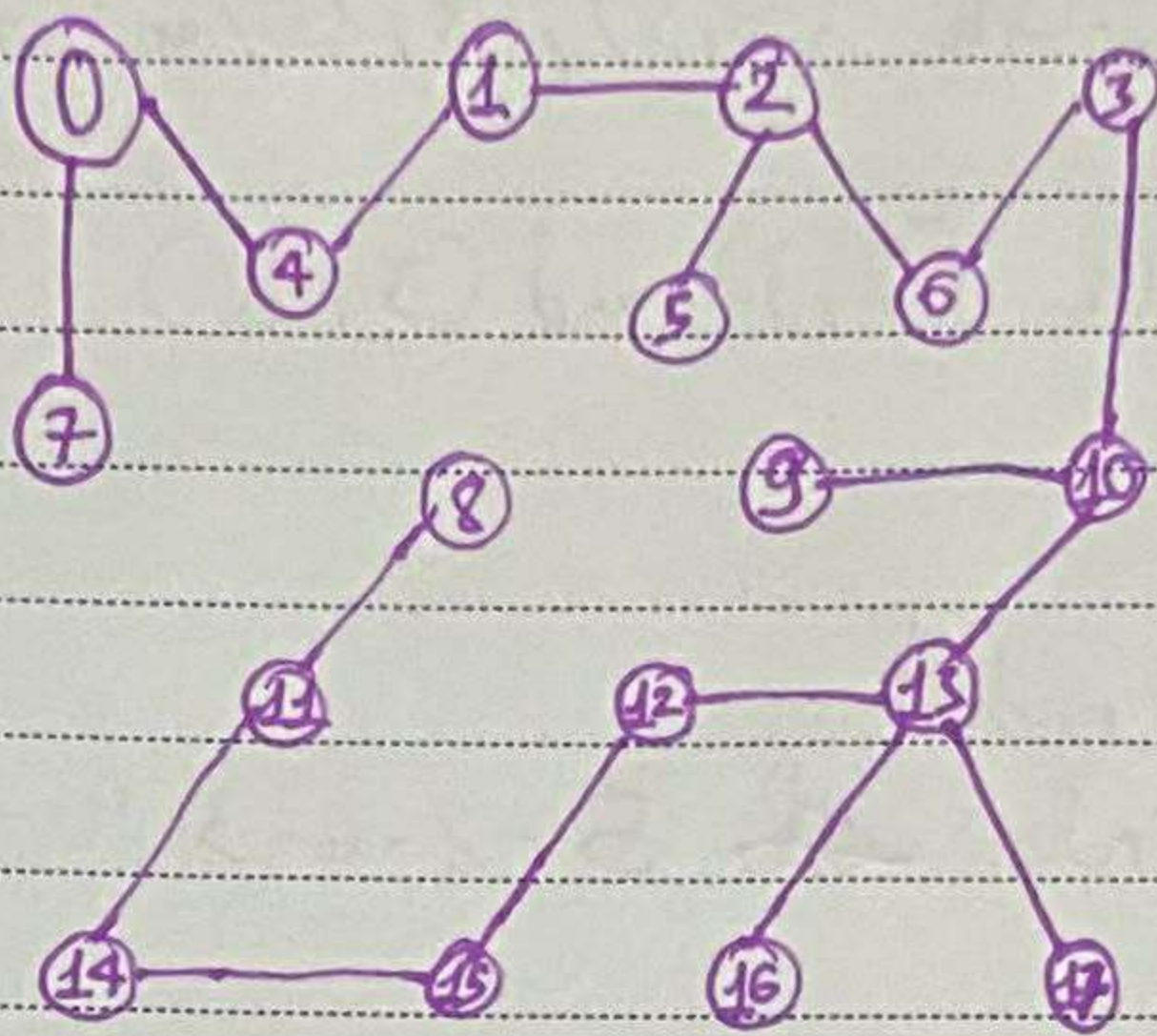
$(12, 15)$, 15 and 6 weight

~~$(15, 14)$, 14 and 1 weight~~

~~$(14, 11)$, 11 and 2 weight~~

$(11, 8)$, 8 and 6 weight, $(13, 17)$ ¹⁷ and 7 weight

⇒ total weight: 64



2. Sort all the ~~add~~ edge in the ascending order, ~~and~~ and we will add the edge if it ~~add a new vertex to the tree~~ doesn't create a circle.

- (1) (0,7) : add 0 and 7
- (1) (12,13) : add 12 and 13
- (1) (14,15) : add 14 and 15
- (1) (13,16) : add 16
- (2) (3,6) : add 6
- (2) (11,14) : add 11
- (3) (2,5) : add 2 and 5
- (4) (3,10) : add 10
- (4) (10,13) : no circle created
- (5) (0,4) : add 4
- (5) (1,4) : add 1
- (5) (1,2) : no circle created
- (5) (9,10) : add 9
- (6) (0,1) : create a circle with (0,4) and (1,4)
→ eliminate
- (6) (2,6) : ~~add~~ no circle created

(6) (6,9) create circle with (6,3), (3,10) and (10,9)
→ eliminate

(6) (6,10) create circle with (6,3) and (3,10)
→ eliminate

(6) (8, 11): add 8

(6) (12, 15): no circle created

(7) (5, 9) : create a circle with 5-2-6-3-10-9-3
→ eliminate

(7) (13, 17): add 17

$\Rightarrow$ The tree has been completed
total weight: 64

